

What happens when one microservice suddenly goes down? Should your entire application crash with it?



What is Resilience?



- Ability of a system to recover from failures.
- Keeps applications available even when services fail.



What is a Circuit Breaker?



- Prevents continuous requests to a failing service.
- Stops cascading failures across the system.
- Improves system stability and response time.



Circuit Breaker States



- Closed → Requests flow normally.
- Open → Requests are blocked because failures exceeded the threshold.
- Half-Open → Allows a few test requests to check if the service has recovered.



What is Retry?



- Automatically retries a failed request.
- Useful for temporary network or service issues.
- Never retry forever.



What is Backoff Strategy?



- Wait before retrying a failed request.
- Prevents overwhelming an already struggling service.



Types of Backoff



- Fixed Backoff
- Linear Backoff
- Exponential Backoff
- Exponential Backoff with Jitter



When NOT to Retry



- Invalid user input (400 Bad Request)
- Authentication/Authorization errors (401/403)
- Business logic failures
- Permanent failures



Popular Libraries



- Opossum (Circuit Breaker)
- Axios Retry
- Polly.js
- Resilience4j (Java ecosystem)

